CHAPTER 3

[Part 4]

PROBABILITY THEORY

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Terminology

Experiment

- A process to yield an outcome.
- Example: Rolling a dice.

Sample space

- A set of all possible outcomes from an experiment.
- For a dice, that would be values 1 to 6.

Element

- An item in the sample space.

Event

- An outcome or combination outcomes from an experiment.
- If you rolled a 4 on the dice, the event is the 4.

How to Compute Probability

- We assign a probability weight $P(x)$ to each element of the sample space.
- Weight represents what we believe to be relative likelihood of that outcome.
- There are two rules in assigning weight:
  i) The weight must be non-negative numbers.
  ii) The sum of the weights of all the elements in a sample space must be 1.

- Let E be an event.
- The probability of *E*, *P(E) is*

$$P(E) = \sum_{x \in E} P(x)$$

- We read this as "*P(E)* equals the sum, over all x such that x is in *E*, of *P(x)*"

Probability Axioms

Let S be a sample space. A probability function P from the set of all events in S to the set of real numbers satisfies the following three axioms:

For all events A and B in S,

1. $0 \le P(A) \le 1$
2. $P(\emptyset) = 0$ and $P(S) = 1$
3. If A and B are disjoint, the $P(A \cup B) = P(A) + P(B)$

A function P that satisfies these axioms is called a probability distribution or a probability measure.

Complementary Probabilities

- The complement of an event A in a sample space S is the set of all outcomes in S except those in A.

$$P(A') = 1 - P(A)$$

- **Example:**

  Let the probability for getting a prize in a lucky draw is 0.075. Thus, the probability of *not* getting a prize in a lucky draw is 1 - 0.075 = 0.925.

The Uniform Probability Distribution

- The probability of an event occurring is:

$$P(E) = \frac{|E|}{|S|}$$

where:

- **E** is the set of desired events.
- **S** is the set of all possible events.
- Note that $0 \leq |E| \leq |S|$:
  - Thus, the probability will always between 0 and 1.
  - An event that will never happen has probability 0.
  - An event that will always happen has probability 1.

Example 1

A coin is flipped four times and the outcome for each flip is recorded.

i) List all the possible outcomes in the sample space.

ii) Find the event (E) that contain only the outcomes in which 1 tails appears.

Solution

- The list of all possible outcomes in the sample space. [heads(H), tails (T)]

HHHH	HHHT	HHTH	
HTHH			
THHH	HHTT	HTTH	HTHT
THHT	TTHH	THTH	HTTT
TTHT	TTTH	THTT	TTTT

- The event E that contains only the outcomes in which 1 tails appears.

E={ HHHT, HHTH, HTHH, THHH }

Example 2

Two fair dice are rolled. Find the event (E) that the sum of the numbers on the dice is 7.

Solution

- The number on top face of each di 1,2,3,4,5,6.
- Let A={1,2,3,4,5,6}
- The sample space is,
- The event E that the sum of the numbers on the dice is 7 is,

$E = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$

Example 3

Suppose that a coin is flipped and a dice is rolled.

i) List the members of the sample space.

ii) List the members of the event "the coin shows a head and the dice shows a number less than 4".

Solution

- The sample space,

  $(H,1)$ $(H,2)$ $(H,3)$ $(H,4)$ $(H,5)$ $(H$

  $(T,1)$ $(T,2)$ $(T,3)$ $(T,4)$ $(T,5)$ $(T,$

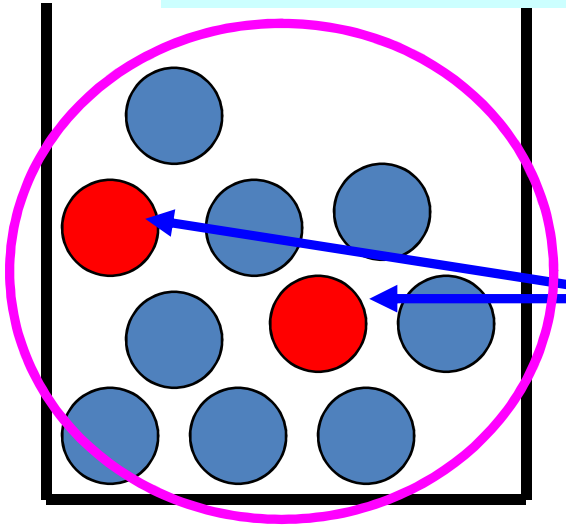
- The members of the event "the shows a head and the die show number less than 4,

  $(H,1)$ $(H,2)$ $(H,3)$

Example 4

- Question: What is the probability of picking a red marble out of a bowl with 2 red and 8 blue?

$$p(\text{red}) = \frac{\text{outcomes in which red occurs}}{\text{total number of outcomes}}$$



There are 2 outcomes that are red

There are 10 total possible outcomes

$$p(\text{red}) = 2 / 10 = 0.2$$

Example 5

Two fair dice are rolled. What is the probability that the sum of the numbers on the dice is 10?

Solution

- The size of sample space is,

  $6 \times 6 = 36$

- 3 possible ways to obtain the sum

  (4,6), (5,5), (6,4)

- The probability is, $\frac{3}{36} = \frac{1}{12}$

Example 6

Five microprocessors are randomly selected from a lot of 1000 microprocessors among which 20 are defective. Find the probability of obtaining no defective microprocessors.

Solution

- There are C(1000,5) ways to select 5 microprocessors among 1000.
- There are C(980,5) ways to select 5 microprocessors since there are 1000-20=980 good microprocessors
- The probability is, $\frac{C(980,5)}{C(1000,5)} = 0.903$

Example 7

There are exactly 3 red balls in a bucket of 15 balls. If we choose 4 balls at random, what is the probability that we do not choose a red ball?

Solution

- There are C(15,4) ways to select 4 b among 15.
- If we do not choose a red ball, the C(12,4) ways to select 4 balls amor remaining 12 balls.
- The probability is, $\frac{C(12,4)}{C(15,4)} = \frac{495}{1365} = \frac{3}{9}$

Example 8

- There are 5 red balls and 4 white bc a box.
- 4 balls are selected at random from these balls.
- Find the probability that 2 of the sel

Solution

- The size of sample space is C(9,4)
- Select 2 red balls, C(5,2) ways
- Select 2 white balls, C(4,2) wc
- The probability is, $\frac{C(5,2).C(4,2)}{C(9,4)} = \frac{(10)(}{126}$

Example 9

Suppose that a die is biased (or loaded) so that the number 2 through 6 are equally likely to appear, but that 1 appears three times as likely as any other number to appear.

Solution

- To model this situation, we should ha

$$P(2) = P(3) = P(4) = P(5) = P(6)$$

$$P(1) = 3P(2)$$



$$P(1)+ P(2)+ P(3)+ P(4)+ P(5)+ P($$
$$3P(2)+P(2)+ P(2)+ P(2)+ P(2)+ P($$
$$8P(2) =8$$

Solution (cont'd)

- Therefore,

  $P(2) = 1/8$

  $P(2) = P(3) = P(4) = P(5) = P(6) = 1$

- and $P(1) = 3/8$

Example 10

Refer to Example 9, what is the probability to obtain an **odd** number?

Solution:

$P(1) + P(3) + P(5) = 3/8 + 1/8 + 1/8 = 5/8$

Example 11

- What is the probability that if a fai tossed 6 times you will get
  - Less than 2 heads
  - At least 2 heads

Solution

- The number of possible outcome is:

  $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$

- Let A be the event that less than 2 H's are observed,

  A= { TTTTTT, HTTTTT, THTTTT, TTHTTT, TTTHTT,
  TTTTHT, TTTTTH }

  Then, $P(A) = 7/64$

- Let B be the event that at least 2 H's are observed,

  $P(B) = 1 - (7/64) = 57/64$

Exercise #1

A bag contains 3 red balls and 3 blue balls. Each red ball is valued as 4 and blue ball as 2.

i) What is probability of getting 4 balls with the total value is 10?

ii) What is probability of getting total value of 6 with at most 3 balls?